

Architecture Documentation

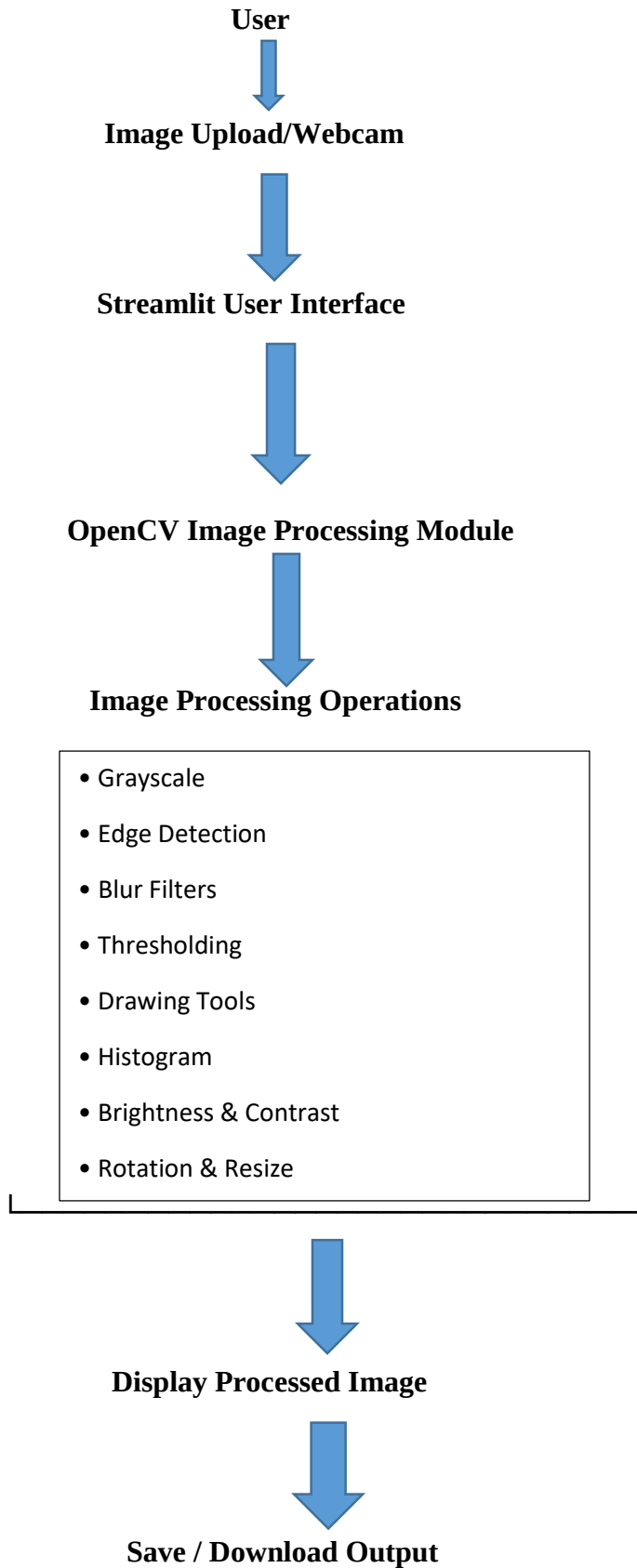
Vision Toolkit Architecture

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Project: Vision Toolkit



ARCHITECTURE EXPLANATION

1. User

The user starts the application and uploads an image or captures one using the webcam.

2. Image Input

The image is loaded into the application using Streamlit. The uploaded image is converted into a NumPy array for OpenCV processing.

3. Streamlit User Interface

The Streamlit interface provides interactive controls that allow users to select different image processing operations and adjust parameters using sliders and buttons.

4. OpenCV Processing Module

OpenCV performs all image processing operations including grayscale conversion, edge detection, thresholding, filtering, brightness adjustment, contrast enhancement, image rotation, resizing, cropping, histogram generation, and drawing functions.

5. Visualization

The processed image is displayed alongside the original image to allow users to compare the results of each operation.

6. Output

The processed image can be saved locally or downloaded through the application.

Technologies Used

- Python 3.14
- Streamlit
- OpenCV
- NumPy
- Matplotlib
- Pillow

Conclusion

The Vision Toolkit follows a modular software architecture where the user interface, image processing logic, drawing functions, webcam module, and utility functions are in different files.